

(1) EM Spectrum

Ranges	γ	X	UV	vis	IR	micro	radio
Energy (E)	$\uparrow$						$\downarrow$
Frequency (ν)	$\uparrow$						$\downarrow$
Wavelength (λ)	$\downarrow$						$\uparrow$
Wavelength (λ)	<0.01	0.01-10	10-400	400-700	700-1e6	1e6-9	>1e9

• violet • indigo • blue • green • yellow • orange • red

(2) $c = \lambda\nu$ (Speed of light = wavelength $\times$ frequency)

c	ν (Hz)	kHz	MHz	GHz
λ (m)	2.9979e8 (m/s)	2.9979e5 (m·kHz)	2.9979e2 (m·MHz)	2.9979e-1 (m·GHz)
nm	2.9979e17 (nm·Hz)	2.9979e14 (nm·kHz)	2.9979e11 (nm·MHz)	2.9979e8 (nm·GHz)

(3) $E = h\nu = hc/\lambda$ (Energy = Planck $\times$ frequency)

hc , h	λ (m)	λ (nm)	ν (Hz)
E (J / photon)	1.9864e-25 (J·m)	1.9864e-16 (J·nm)	6.626e-34 (J·s)
J / mol	1.1962e-1 (J·m/mol)	1.1962e8 (J·nm/mol)	3.9902e-10 (J·s/mol)
kJ / mol	1.1962e-4 (kJ·m/mol)	1.1962e5 (kJ·nm/mol)	3.9902e-13 (kJ·s/mol)

• Grams $\rightarrow$ mol $\rightarrow \times 6.022e23$ for photon count

(4) Quantum Principles

Maxwell	Light is electromagnetic radiation
Planck	Energy emitted only in fixed amounts (quanta) (black-body radiation)
Einstein	Different light has different amounts of fixed energy (photons) (photoelectric effect)
Bohr	Electrons are quantum (line spectra); Rydberg eq.
De Broglie	Electrons are particle and wave (5)
Schrodinger	Wave equation (n , l , m_l , m_s)
Heisenberg	Position and velocity not both knowable

$$E = h\nu = hc/\lambda = R(1/n_f^2 - 1/n_i^2)$$

$$n_i = \sqrt{1/(1/n_f^2 - hc/(R\lambda))}; \quad n_f = \sqrt{1/(1/n_i^2 + hc/(R\lambda))}$$

Variable	Value	Variable	Value
h	6.626e-34 J·s	h/R	3.0394e-16 s
c	2.9979e8 m/s	hc	1.9864e-25 J·m
R	2.18e-18 J	hc/R	9.1120e-8 m

(5) $\lambda = h/(m \cdot u)$ (De Broglie)

- 1 nm = 1e-9 m
- 1 e $^-$ = 9.11e-31 kg

h	u (m/s)	km/s
m (kg)	6.626e-34 (J·s)	6.626e-37 (kJ·s)
g	6.626e-31 (mJ·s)	6.626e-34 (J·s)
amu	3.9892e-7 (J·s·amu/kg)	3.9892e-10 (kJ·s·amu/kg)

(6) Electron Configuration

Symbol	Name	Value
n	Principal energy level (shell)	(1, 2, 3, ...)
l	Sublevel = subshell (shape)	(0, ..., $n - 1$)
m_l	Orbital (orientations)	(- l , ..., l)
m_s	Spin (2 per orbital)	$\pm 1/2$

	l	m_l	# orbitals	# e $^-$
s	0	0	1	2
p	1	± 1	3	6
d	2	± 2	5	10
f	3	± 3	7	14

- 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p 7s 5f 6d 7p
- Cr [Ar]4s¹3d⁵;
- Cu [Ar]4s¹3d¹⁰;

(7) Reading EC

1. Add every superscript to get Z

Para / diamagnetic

1. Draw boxes for the outer shell
2. Fill singly, parallel spins, before pairing (Hund)
3. If any unpaired e $^-$ then paramagnetic, else diamagnetic

Excited / ground state

- Ground = Follows Aufbau
- Excited = an e $^-$ sits higher than Aufbau

Valence electrons

- Count e $^-$ in the **highest** n (outer s + p) = the group number, 1A $\rightarrow$ 1 ... 8A $\rightarrow$ 8
- Everything below the highest n is **core**

(8) Ion Configuration

1. Start from the neutral atom
2. **Anion**: add e $^-$ into the outer p until it reaches a noble-gas configuration
3. **Cation**: remove from the **highest** n **first**
 - Transition metals lose **4s before 3d**
 - Exam ions are main-group, not transition metals

(9) Quantum Numbers, Penetration & Shielding

- Shell n holds n^2 orbitals, $2n^2$ e $^-$
- **energy**: **s < p < d < f** (within shell, same n)
- **Pauli exclusion** — no two e $^-$ carry the same four quantum numbers $\rightarrow$ 2 per orbital, opposite spins
- **Penetration** — orbital reaching into the core region
- $\uparrow$ shielding $\rightleftharpoons \uparrow$ reduced Z_{eff} $\rightleftharpoons \downarrow$ penetration $\rightleftharpoons \uparrow$ **energy** (reverse also holds)
- **Shielding** — only **core** (inner) e $^-$ shield effectively; valence e $^-$ shield each other poorly

(10) Z_{eff} & Core Electrons

$$Z_{eff} \approx Z - (\text{core } e^-)$$

1. Core e $^-$ = total e $^-$ — valence e $^-$ (= the **previous noble gas**)
2. $Z_{eff} \approx Z - \text{core } e^-$
 - Same period $\rightarrow$ same core, so the **rightmost** element wins
 - Poor valence shielding is why Z_{eff} rises across a period

(11)–(13) Trends

Trend	$\rightarrow$ across period	$\downarrow$ down group
Z_{eff}	increases	$\sim$ constant
Atomic radius	decreases	increases
Ionization energy	increases	decreases
Electron affinity	more favorable	less favorable

Atomic size

- Down a group **bigger** (new shell) • across a period **smaller** (Z_{eff} pulls in)
- $\uparrow$ = **lower-left** • $\downarrow$ = **upper-right**
- Group position beats period position when both apply
- cation < neutral < anion

Ionization energy

- $\uparrow$ = **upper-right** (He $\uparrow$ of all) • $\downarrow$ = **lower-left** (opposite of radius)
- $IE_2 > IE_1$ always
- **Big jump** appears once the first **core** e $^-$ is removed

Electron affinity

- **Most eager for an e $^-$ = halogens, upper right** — this course writes it both as "most negative EA " and "largest positive EA "; either phrasing means the same element
- Noble gases and full/half-full subshells $\approx$ 0 or unfavorable
- **EA (X) and IE of X $^+$ are the same magnitude** — F + e $^-$ releases exactly what F $\rightarrow$ F costs

(14) Families & Diagonal Rule

1A	alkali metals	most reactive metals; ↑ down
2A	alkaline earth	reactive, less than 1A
7A	halogens	reactive nonmetals; ↓ down
8A	noble gases	inert — full octet

- **Diagonal rule** — an element resembles its down-right neighbor: Li~Mg · Be~Al · B~Si

HW-only gaps — off the exam list

Oxides — metal oxide = **basic** (K_2O , MgO , Fe_2O_3 , Cr_2O_3) · nonmetal oxide = **acidic** (P_4O_{10} , SO_2 , CO_2 , Cl_2O_7) · Al_2O_3 and BeO = **amphoteric**

Formula from charges — 1A 1+ · 2A 2+ · 3A 3+ · 5A 3- · 6A 2- · 7A 1-; criss-cross and reduce

• Same-group swap keeps the pattern: $BaS \rightarrow CaO$

• $2A + 3F_2 \rightarrow 2AF_3 \Rightarrow A$ is 3+ (**Al**) · $3Li + Z \rightarrow Li_3Z \Rightarrow Z$ is 3-, so Mg gives **Mg₃Z₂**

Bond-type ID — ionic **and** covalent together = a salt of a **polyatomic ion** (NH_4NO_3 , K_2SO_4); all-nonmetal = covalent only

Family reactions

• $2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$

• $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$

Odds & ends

• Lanthanides = 4f row (**Ce**) · actinides = 5f (U)

• Diatomic: H_2 N_2 O_2 F_2 Cl_2 Br_2 I_2

• Metallic character ↑ down, ↓ across → **lower-left** wins

• Isoelectronic = matches a noble gas e⁻ count (K^+ , Cl^- , Ca^{2+} = Ar)

(15) Lattice Energy

Energy **released** when gaseous ions form the solid crystal.

$$U \propto \frac{q_1 q_2}{r_1 + r_2}$$

1. Compare charges first

2. If charges tie, smaller ions win

• ↑ charge → ↑ **U** (dominant) · ↑ ionic radius → ↓ U

(16) Electronegativity

EN = an atom's ability to attract the electrons shared in a bond.

$$\Delta EN = |EN_A - EN_B|$$

• Read **EN** off the chart — **EN** ↑ up and to the right, F highest, noble gases excluded

(17) Bond Polarity

1. Take the absolute difference (ΔEN)

2. Classify

~0	nonpolar covalent
0.4 – 1.9	polar covalent
≥ 2.0	ionic

• **"Most polar"** → ΔEN **alone** (largest difference wins)

• **"Strongest / shortest bond"** → **bond order FIRST**: triple > double > single. ΔEN only breaks ties between equal orders — a $C \equiv N$ beats a high- ΔEN $Si=O$

• Larger ΔEN → stronger, shorter bond, but the guide flags this as *not always true*

(18) Formal Charge & Resonance

$$FC = (\text{valence } e^-) - [\text{nonbonding } e^- + \# \text{ bonds}]$$

1. **FC** on every atom

2. Check ΣFC = the overall charge

3. Pick the structure with **FC's nearest 0**

4. Put any -1 on the **more electronegative** atom

• Equivalent resonance forms → real bonds are the **average** (e.g. $1\frac{1}{2}$)

(19) Lewis Structures x4

1. Total valence e⁻ — +1 per negative charge, -1 per positive
2. Central atom = least electronegative, never H
3. Single bonds out to every atom
4. Fill outer octets with lone pairs (H takes 2)
5. Leftover e⁻ onto the central atom
6. Central below 8 → pull an outer lone pair into a double/triple bond, **unless** Al/B/Be
7. Check formal charges — see (18)

Four structures — one of each type:

(a) Normal octet	8 valence e ⁻ per atom (H wants 2) — H_2O , CH_4 , NH_3 , CO_2 , CH_2Cl_2 , CH_3CH_2Cl
(b) Incomplete octet	Al, B, Be only — content below 8, do not add double bonds to fix. BF_3 / BCl_3 / AlI_3 leave 6 e ⁻ on the center; $BeCl_2$ leaves 4
(c) Expanded octet	Central atom period 3 or below (S, P, Cl, Xe, As, I) uses empty d orbitals for >8 — $AsCl_5$, XeI_2 , SO_4^{2-}
(d) Organic	-OH alcohol · C=O aldehyde/ketone · -COOH carboxylic acid · -NH₂ amine

- Every C makes 4 bonds; a **-COOH** carbon carries one **=O** and one **-O-H**; every O gets 2 lone pairs
- Ions go in brackets with the charge outside
- If the all-single-bond form leaves the center with a nonzero **FC**, add double bonds to zero it (e.g. two S=O in SO_4^{2-})

(20) Photoelectric Effect ($h\nu$ - binding / work)

$$KE = h\nu - \Phi = h(c/\lambda) - \Phi = \frac{1}{2}m_e u^2$$

$$u = \sqrt{(2hc/m_e)(1/\lambda - 1/\lambda_{thr})}$$

Variable	Value
h	6.626e-34 J·s
m_e	9.11e-31 kg
hc	1.9864e-25 J·m
$2hc/m_e$	4.3609e5 m ³ /s ²

Have	Want	Use
ν_{thr} or λ_{thr}	Φ	$\Phi = h\nu_{thr} = h(c/\lambda_{thr})$
Φ	λ_{thr} (max ej. λ)	$\lambda_{thr} = hc/\Phi$
Φ	Φ per mol	$\Phi \times N_A$ (/1e3 for kJ/mol)
Φ , λ	u	1. $KE = h(c/\lambda) - \Phi$ 2. $u = \sqrt{2KE/m_e}$
λ , λ_{thr}	u	2nd eq above (skips Φ and KE)

(21) Heisenberg Uncertainty

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}, \quad \Delta p = m\Delta u$$

1. $\Delta p = m \cdot \Delta u$ (kg, m/s)

2. $\Delta x \geq h/(4\pi \cdot \Delta p)$

• Result is the **minimum** uncertainty

• Rearrange the same way to solve for Δu

• Keep everything in kg, m, s

(22) Born-Haber Cycle

$$\Delta H^\circ_f = \Delta H_{sub} + \Sigma IE + \frac{1}{2}BE + EA + U$$

1. List every step with its sign: sublimation (+), **IE** (+), bond dissociation (+), **EA** (usually -), **U** (-)

2. **Halve** the bond dissociation if you need only one atom

3. Chain: elements → gaseous atoms → gaseous ions → solid

4. Set the chain equal to ΔH°_f

5. Solve for the single unknown

• Use **IE₂** as well when the cation is 2+

• U runs ions(g) → solid, so it comes out **negative**; report |U| if asked for a positive lattice energy

(23) Bond Enthalpy

$$\Delta H_{rxn} = \Sigma \Delta H(\text{bonds broken}) - \Sigma \Delta H(\text{bonds formed})$$

1. List bonds broken (reactants) and formed (products), using the balanced coefficients

2. Σ BE(broken), always positive

3. Σ BE(formed), subtracted

4. $\Delta H = \Sigma$ broken - Σ formed

• Bonds unchanged on both sides cancel — skip them

• Negative ΔH = exothermic

• For "heat released by g of X," convert g → mol and scale